

WEEDON FIELD, AL, USA

WMO: 720363

Lat: 31.951N

Lon: 85.129W

Elev: 87

StdP: 100.29

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.2	(c) -2.2	(d) -12.5	(e) 1.3	(f) 2.3	(g) -10.0	(h) 1.6	(i) 3.0	(j) 7.4	(k) 15.6	(l) 6.4	(m) 15.0	(n) 0.9	(o) 340	(p) 0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	36.6	24.4	35.6	24.5	34.6	24.5	26.8	33.1	26.3	32.5	25.8	31.8	2.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.6	28.9	24.7	19.9	28.6	24.2	19.3	28.3	84.0	33.1	81.9	32.5	79.9	32.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.2	4.6	-7.1	38.1	2.0	1.6	-8.5	39.2	-9.7	40.1	-10.8	41.0	-12.3	42.2
			-8.1	27.8	2.0	0.6	-9.6	28.3	-10.8	28.6	-11.9	29.0	-13.4	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	9.3	11.8	15.3	18.9	23.2	27.2	28.0	28.0	25.4	19.6	13.7	11.4
	DBStd	7.77	5.61	5.32	4.80	3.64	3.07	1.85	1.59	1.77	2.72	4.35	4.58	5.44
	HDD10.0	201	83	40	11	1	0	0	0	0	0	1	16	49
	HDD18.3	1047	284	191	118	35	6	0	0	0	1	38	152	222
	CDD10.0	3618	60	90	175	269	411	516	558	557	463	299	127	94
	CDD18.3	1423	3	7	24	53	158	266	300	298	213	78	13	9
	CDH23.3	14416	16	58	261	636	1627	2786	3121	3072	1948	745	106	41
	CDH26.7	6803	0	8	59	195	725	1476	1598	1549	918	253	16	5
Wind	WSAvg	1.6	1.8	1.9	2.1	1.9	1.6	1.4	1.3	1.2	1.3	1.4	1.4	1.6
Precipitation	PrecAvg	1396	130	116	161	101	84	136	160	113	99	70	118	101
	PrecMax	1708	306	236	322	198	144	261	511	236	339	178	295	187
	PrecMin	1043	28	29	6	37	6	51	63	42	30	1	33	48
	PrecStd	190	64	54	78	46	36	64	99	45	79	54	71	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	28.1	30.4	32.2	35.5	38.1	37.4	38.2	36.3	32.9	28.7	27.0
		MCWB	18.9	20.2	18.8	20.5	21.7	23.8	25.2	25.3	23.5	22.6	20.3	19.9
	2%	DB	22.9	25.1	28.4	30.5	33.5	36.4	36.0	36.4	34.7	31.3	26.1	24.2
		MCWB	17.8	17.9	18.7	19.8	22.0	24.1	25.2	24.9	23.4	21.6	18.7	18.4
	5%	DB	20.6	22.9	26.4	29.0	32.1	35.2	34.9	35.1	33.3	29.6	24.3	22.0
		MCWB	17.0	16.5	17.9	19.2	21.7	24.4	25.2	25.1	23.6	20.8	17.5	18.4
	10%	DB	18.5	20.8	24.2	27.4	30.8	33.8	33.8	33.8	31.9	27.8	22.3	20.1
		MCWB	15.3	16.0	16.2	18.2	21.3	24.1	25.0	25.0	23.6	19.8	17.0	17.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	21.4	21.0	23.2	24.8	26.8	27.3	27.5	26.3	25.4	23.1	21.8
		MCDB	22.5	25.9	25.5	28.3	29.4	33.3	33.5	34.0	31.6	28.7	24.8	23.8
	2%	WB	19.1	19.7	20.1	21.9	23.9	26.1	26.7	26.8	25.6	24.1	21.3	20.4
		MCDB	21.5	23.1	25.1	26.9	29.1	33.0	33.1	33.1	31.1	27.9	23.7	22.3
	5%	WB	17.8	18.6	19.2	20.9	23.2	25.6	26.2	26.3	25.1	23.0	19.7	19.2
		MCDB	20.2	21.7	24.3	25.6	28.8	32.3	32.7	32.4	30.5	26.7	22.7	21.3
	10%	WB	16.2	17.0	18.1	20.0	22.6	25.1	25.7	25.8	24.5	21.9	18.1	17.9
		MCDB	18.4	19.9	22.8	24.7	28.3	31.5	31.8	31.5	29.2	25.5	21.2	20.1
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.5	14.3	14.5	14.0	12.7	11.8	11.5	12.2	14.1	14.5	12.1
		MCDBR	13.4	15.0	15.9	16.4	15.6	14.8	13.3	13.7	13.9	15.0	15.7	13.3
		MCWBR	8.7	8.5	7.4	7.0	6.0	4.8	4.5	4.5	5.1	6.8	8.8	8.3
	5% WB	MCDBR	11.3	11.9	13.2	12.2	12.4	12.6	11.8	11.5	11.5	11.1	11.4	10.3
		MCWBR	8.3	8.1	6.9	6.4	5.3	4.9	4.5	4.4	4.8	5.7	7.3	7.2
Clear-Sky Solar Irradiance	taub	0.326	0.329	0.355	0.387	0.435	0.480	0.516	0.512	0.427	0.373	0.350	0.329	
	taud	2.522	2.509	2.448	2.354	2.257	2.190	2.097	2.103	2.329	2.454	2.464	2.527	
	Ebn at Noon	900	931	926	904	858	816	784	782	842	869	861	875	
	Edh at Noon	86	95	109	124	138	147	161	158	121	100	90	81	
All-Sky Solar Radiation	RadAvg	2.91	3.48	4.64	5.75	6.33	6.12	6.03	5.53	4.92	4.22	3.27	2.51	
	RadStd	0.17	0.42	0.39	0.44	0.47	0.47	0.37	0.34	0.41	0.49	0.28	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+109

Nomenclature: See separate page